36 Reactions on surfaces and Review of catalysis

This lecture is based on the [Nobel Lecture by Gerhard Ertl]

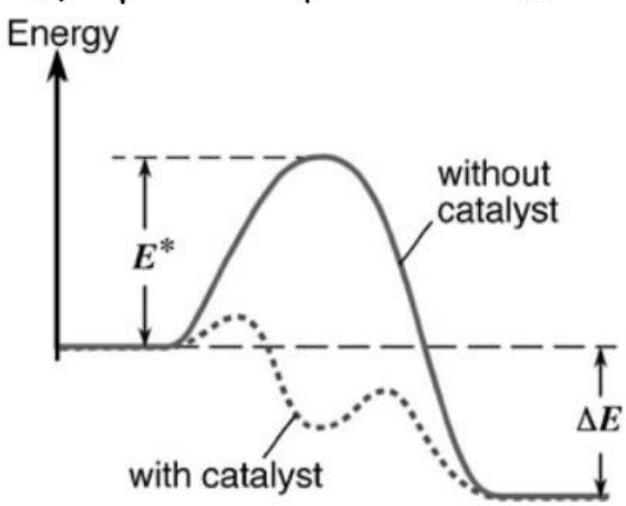
A chemical reaction involves breaking of existing chemical bonds & formation of new chemical bonds

This involves transfer of energy from the potential energy stored w/in the original chemical bonds. For example, some of this energy may be released as heat (exothermic rxn)

For the reaction to occur, enough energy must be available (or supplied) to overcome the activation energy, ΔE^{act}

The rate is proportional to $e^{\frac{-\Delta E^{act}}{RT}}$; hence, the rate can be increased by decreasing ΔE^{act}, increasing T

A catalyst decreases the activation energy by enabling an alternate reaction pathway. This new pathway results in the formation of new intermediate species & is typically a lower energy pathway than the uncatalyzed reaction



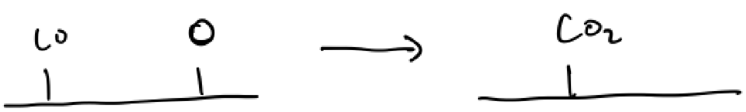
In catalysis, the reaction happens on the surface

This happens in several steps:

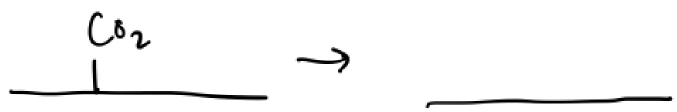
1. Reactant molecules "adsorb" to the catalyst surface. Adsorption is formation of a bond btwn the reactant molecule & the surface. Bond can be either chemical or physical.

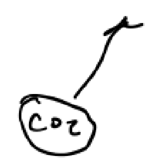
2. Reactant molecules fully or partially decompose on the catalyst surface. For example:

3. Atoms and/or molecules on the catalyst surface react with each other. For example:



4. Product molecules "desorb" from the catalyst surface. Desorption is the reverse of adsorption. For example:





The catalyst surface readily binds with atoms and molecules because the atoms in the surface are chemically "unsaturated" due to creation of the surface. They are "hungry" to form new bonds.

An important application of catalysis is the ammonia synthesis reaction, N2 + 3H2 --> 2NH3. In the first part of the 20th century, the world was facing starvation due to an increasing population and depletion of fertilizers. The Haber-Bosch process was designed in the 1930s to combat this by generate ammonia from air.

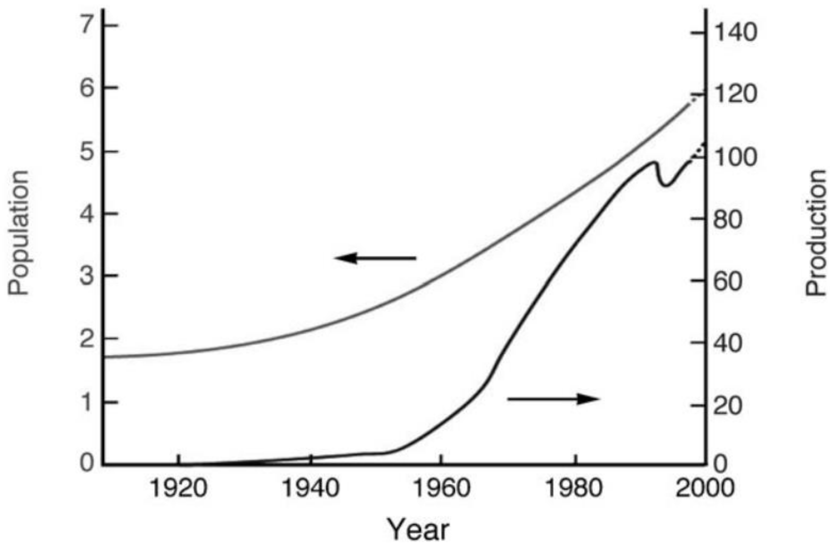


Figure 3. Variation of the world population (in 10^9) and the ammonia production (in 10^6 metric tons of nitrogen) during the 20th century.[7]

The Haber-Bosch process utilizes catalysts comprised of iron

Industrial ammonia synthesis catalysts comprise nanometer sized particles with surface areas of ~ $20 \frac{m^2}{g}$

Iron oxide present under reaction conditions is reduced to metallic iron by incorporating potassium oxide

The particles are stabilized against agglomeration by attaching to an Al_2O_3 support

The main utility of the Fe catalyst is to dissociate N_2

On the catalyst surface, this looks like:

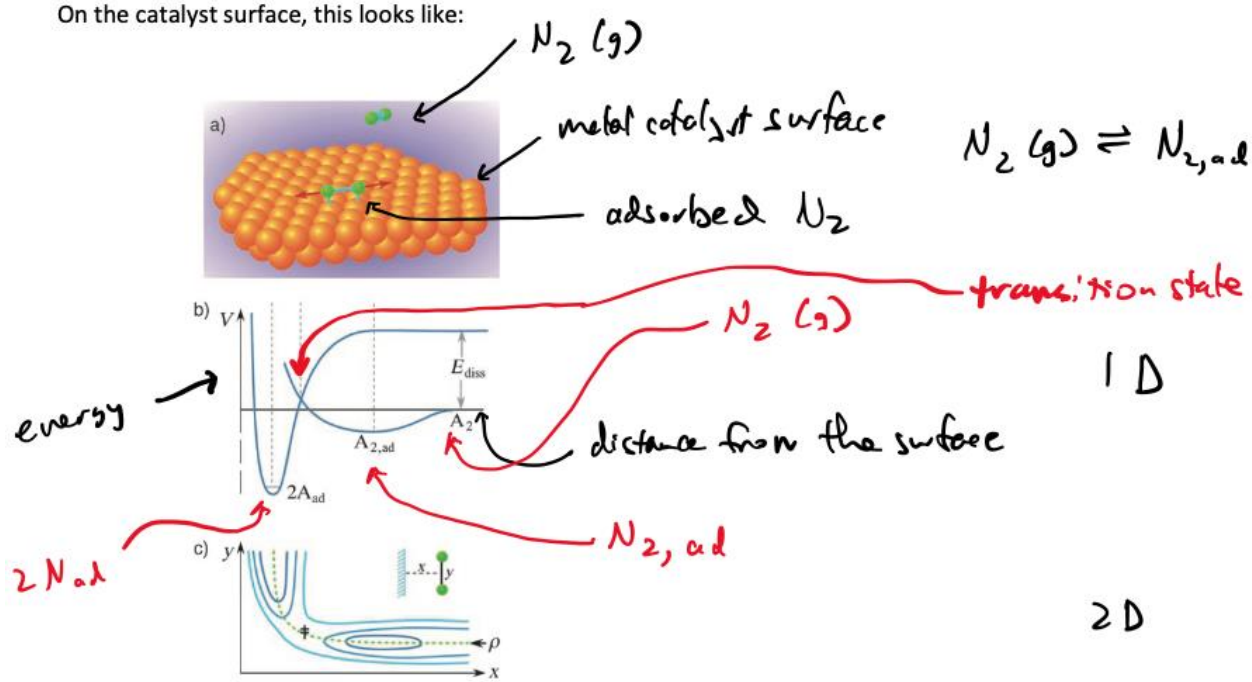


Figure 5. a) Illustration of the process of dissociative chemisorption of a diatomic molecule; b) one-dimensional Lennard–Jones diagram; c) two-dimensional representation of equipotential lines as function of the distance *x* of the molecule from the surface and the separation *y* between the two atoms.

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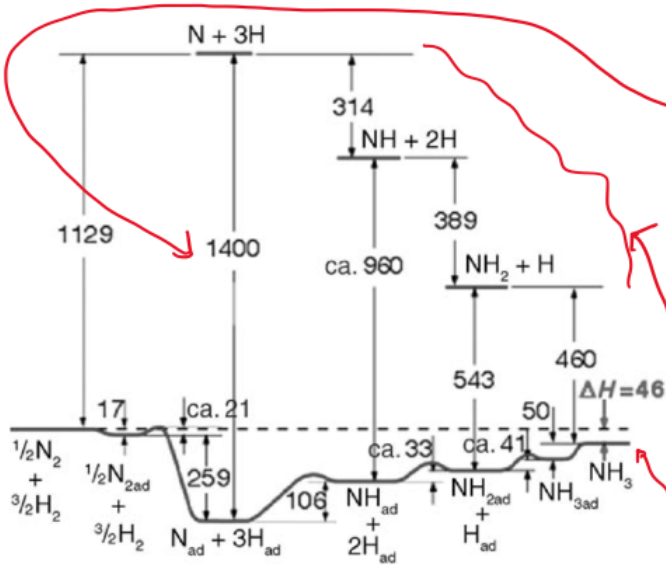
energies in kJ/mol of gas phase vs catalyzed reaction pathways



energy of N (g) + 3H (g) vs N_{ad} + $3H_{ad}$. The difference is 1400 $\frac{kJ}{mol}$

gas phase pathway

catalytic pathway

the catalyst provides a lower energy pathway

From this diagram, we can write the mechanism for the Fe-catalyzed reaction:

$N_2 (g) \rightleftharpoons N_{2,ad}$

$H_2 (g) \rightleftharpoons H_{2,ad}$

$N_{2,ad} \rightleftharpoons 2N_{ad}$

$H_{2,ad} \rightleftharpoons 2H_{ad}$

$N_{ad} + H_{ad} \rightleftharpoons NH_{ad}$

using * notation

$N_2 (g) + * \rightleftharpoons N_2*$

$H_2 (g) + * \rightleftharpoons H_2*$

$N_2* + * \rightleftharpoons 2N*$

$H_2* + * \rightleftharpoons 2H*$

$N* + H* \rightleftharpoons NH* + *$

$NH* + H* \rightleftharpoons NH_2* + *$

$NH_{ad} + H_{ad} \rightleftharpoons NH_{2,ad}$

$NH_{2,ad} + H_{ad} \rightleftharpoons NH_{3,ad}$

$NH_{3,ad} \rightleftharpoons NH_3\ (g)$

$NH_2* + H* \rightleftharpoons NH_3* + *$

$NH_3* \rightleftharpoons NH_3\ (g) + *$

*'d species occupy a catalyst site (i.e., are adsorbed to the catalyst), * are empty catalyst sites

In this notation, "ad" means that the atom or molecule is

adsorbed to the catalyst